LAWRENCE
LIVERMORE
NATIONAL
LABORATORY

LLNL-SR-854547

US Nuclear Testing: Health Consequences and Policy Decisions

G. Islas

September 19, 2023

Disclaimer

This document was prepared as an account of work sponsored by an agency of the United States government. Neither the United States government nor Lawrence Livermore National Security, LLC, nor any of their employees makes any warranty, expressed or implied, or assumes any legal liability or responsibility for the accuracy, completeness, or usefulness of any information, apparatus, product, or process disclosed, or represents that its use would not infringe privately owned rights. Reference herein to any specific commercial product, process, or service by trade name, trademark, manufacturer, or otherwise does not necessarily constitute or imply its endorsement, recommendation, or favoring by the United States government or Lawrence Livermore National Security, LLC. The views and opinions of authors expressed herein do not necessarily state or reflect those of the United States government or Lawrence Livermore National Security, LLC, and shall not be used for advertising or product endorsement purposes.

This work performed under the auspices of the U.S. Department of Energy by Lawrence Livermore National Laboratory under Contract DE-AC52-07NA27344.

US Nuclear Testing: Health Consequences and Policy Decisions

Introduction:

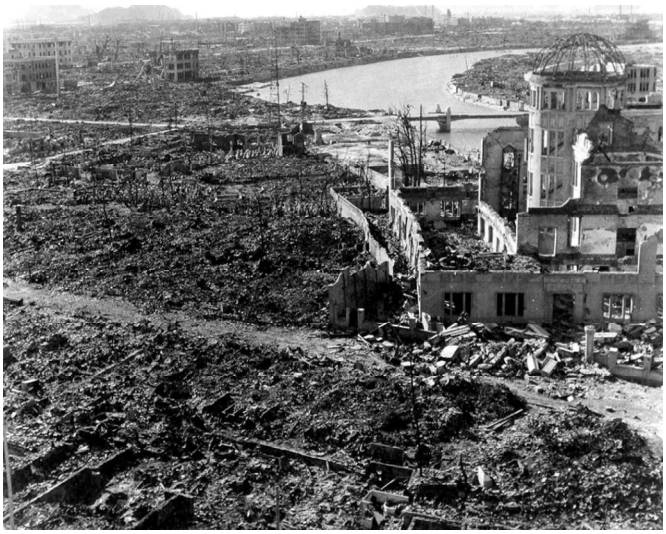
With the approval of President Franklin D. Roosevelt to begin the research on a nuclear bomb in 1941 to the last test conducted by the U.S. in 1992, the fifty-year history of the nuclear weapons testing program has been an expansive topic of research. The program's growth is credited to the race to build the first atomic weapon for war. The termination of the Soviet Union and many other factors, including concerns about the adverse health effects of radioactive fallout, influenced the decline of the need for the program. The discovery of the negative health effects caused by low-level radiation and the subsequent studies influenced sitting U.S. presidents in passing policies that significantly impacted the nuclear weapons testing program.

The Bomb and Testing:

In the early 1940s, the military and scientists from all over the nation were in the race of their lives to build the first atomic bomb. In 1941, Vannevar Bush, the Office of Scientific Research and Development director, informed President Franklin D. Roosevelt about the ability and need to produce the first atomic bomb that would affect World War II. The president then approved him to begin the research and production of the first nuclear weapon used for war. The Manhattan Project, overseen by General Leslie R. Groves and scientifically carried out by J. Robert Oppenheimer, took place from 1942 to 1945 and was comprised of work from scientists all over the nation. On July 16th, 1945, *Trinity*, the first atomic bomb, was tested on the Alamogordo Bombing Range in New Mexico. While this test would later significantly affect its

use in the war and the future of the nuclear testing program, other concerns later arose, such as the health effects created by the fallout emitted from these bombs. The fallout of a nuclear weapon is the radioactive particles that fall to Earth due to a nuclear explosion.

After creating the first atomic bomb and the *Trinity* test, the U.S. used the nuclear weapon during World War II on Hiroshima and Nagasaki. The damage from these bombs was calamitous, resulting in the deaths of over 100,000 people instantly. The amount of power within these bombs led scientists and government officials to question themselves and their morals on the future uses of this weapon and the nuclear weapons testing program. Some scientists brought up questions of ethics and personal morals regarding the matter. After the bombings in Hiroshima and Nagasaki, J. Robert Oppenheimer expressed his desire to ban these nuclear weapons. At the same time, other scientists, such as Edward Teller, saw the bombing as another reason to continue further research on nuclear weapons.



"Hiroshima and Nagasaki: 75th Anniversary of Atomic Bombings." *BBC News*, 8 Aug. 2020, www.bbc.com/news/in-pictures-53648572.

After further debate, President Truman approved continuing the research of the nuclear weapons testing program. The president agreed with Edward Teller and other government officials and decided the research would be essential for its part during the Cold War. In 1952, the first hydrogen bomb was created and tested by Teller and other scientists. The hydrogen bomb was said to have the potential to be 1,000 times more powerful than the atomic bomb. Although no country has used the hydrogen bomb during battle, the testing of the thermonuclear bomb still caused immense damage to the surrounding areas.

Most of the atmospheric nuclear weapon tests took place at the Nevada Test Site. After many studies and reports of health concerns surrounding the area, scientists urged the halting of atmospheric, underwater, and space testing, and many of the general public protested around the Nevada Test Site in favor of the ban. President Eisenhower was interested in halting the nuclear tests. However, he had a divided administration, with top advisors at the Atomic Energy Commission and the Pentagon supporting the nuclear testing and downplaying the dangers of fallout. (Evans, (n.d))

Lane, Taylor. “Nevada National Security Site through the Years - Photos.” *Journal*, 17 Aug. 2022, www.reviewjournal.com/local/local-nevada/nevada-national-security-site-through-the-years-photos-2624752/.

Dangers and People:

As mentioned before, scientists investigated health concerns from the fallout emitted from these nuclear weapons. The effects of radiation on the human body can be detrimental. "Ionizing radiation is a carcinogenic agent that has been proven to cause cancer leukemia in humans. Radiation can also cause genetic damage when its swift-moving energy bombards our cells and tissues." (Ball).

In the following years after the Hiroshima and Nagasaki bombings, scientists in the Atomic Bomb Casualty Commission unveiled the health effects of radiation from the bombs after multiple studies conducted on the surrounding areas and populations within the Nevada Test Site and other affected countries. After the U.S. conducted an atmospheric nuclear weapons test, regions surrounding the testing site would be exposed to the radiation emitted. The families living in Arizona, Nevada, and Utah were exposed to most of the fallout and were known as *Downwinders*. The fallout created by Trinity caused harm to livestock living in the ranches near the facility in New Mexico, and families received significant exposure to fallout for the next two weeks. Soon after these tests, scientists conducted multiple studies to monitor the health effects that soon followed.

As noted, the instant damage of the bombings on Hiroshima and Nagasaki was catastrophic. With over 100,000 deaths instantaneously, many more would soon follow due to the amount of radiation the survivors of the bomb were exposed to. By the end of 1945, news noted that the number of deaths had doubled from the effects of the bombings, resulting in a total of about 200,000 deaths. After subsequent studies, scientists understood radiation's genetic influences on the human body. "Though exposure to radiation can cause acute, near-immediate effect by killing cells and directly damaging tissue, radiation can also have effects that happen on

longer scale, such as cancer, by causing mutations in the DNA of living cells." (Listwa) After the bombings, there was an increase in Leukemia among the atomic bomb survivors. The growth came two years after the bombings and would peak around four to six years later. An increase in other cancers would later be noted around ten years after the bombings when tumor registries commenced in Hiroshima and Nagasaki in 1956.

While many scientists were trying to warn government officials about the dangers of radioactive fallout, some citizens of the United States were also trying to bring awareness to the situation. After hearing about the damage caused to the survivors of the Hiroshima bombing, Norman Cousins, an editor of the *Saturday Review* and peace activist, helped bring 25 female Hiroshima survivors who had suffered scars and deformities to the U.S. for treatment. They were later called "the Hiroshima Maidens."

> "The danger of fallout exposure received wide publicity as Hiroshima victims ("the Hiroshima Maidens") visited the United States for treatment. Especially prominent was the case of the Japanese fishing ship Lucky Dragon whose crew encountered fallout produced by the fifteen megaton "Bravo" shot on 28 February 1954. Radiation exposure certainly played a role in the death of one crew member" (Evans, (n.d)).

As mentioned before, another instance where the fallout from a nuclear weapon was detrimental to public health was in 1954. On the morning of March 1st, 1954, at 6:45 am (February 28th, at 6:45 pm GMT), the U.S. tested a nuclear weapon, *Bravo*, shot from the Marshall Islands. *Bravo* was supposed to be the first of many thermonuclear tests of *Operation Castle*. "The explosion was more than two and a half times greater than expected and caused far higher levels of fallout and damage than scientists had predicted." (Atomic Heritage Foundation)

In the following hours of the explosion, areas and citizens surrounding the test site would be exposed to the fallout, which rained down as a white, powder-like substance. The Japanese fishing boat, *Lucky Dragon*, was among the many areas and people exposed to the radiation. Because of the drastic effects of the test, the urgency to shut down atmospheric testing became a primary publicized concern from the general public. "It wasn't the World War II bombings but the *Lucky Dragon No. 5* that finally triggered the world-wide call to ban the bomb, the first truly global protest movement of the modern era." (Ropeik).

After the *Lucky Dragon* incident, the general public and scientists from the test sites called for an end to atmospheric tests due to the health effects caused by the radioactive fallout. The U.S. conducted the majority of nuclear weapon tests at the Nevada Test Site. Although the site was very rural and flat, ideal for the tests and security, government officials did not adequately warn many families living nearby of the effects soon to come. As mentioned before, the families living in these surrounding areas were known as *Downwinders*, and because "many of the communities in the surrounding ranch towns were subsistence farmers, and consumed locally produced milk, beef, and vegetables, all of which were exposed to fallout from the tests," (AHF), this made the *Downwinders* highly vulnerable to radiation exposure.

"Downwinders and the Radioactive West." *PBS*, 3 Oct. 2022, www.pbs.org/video/downwinders-and-the-radioactive-west-usugap/.

One of the most concerning *Downwinders* cases was after a nuclear weapon test, later nicknamed "Dirty Harry." The U.S. conducted the test on May 19th, 1953, as part of *Operation Upshot-Knothole*. Once again, the detonation of a nuclear weapons test surpassed the scientists who designed it, as it yielded twenty kilotons over the advisable yield level to limit radiation exposure. The fallout from the bomb was then carried over to the northeast side of St. George, Utah, by strong winds. As the public in the area was warned and advised to take shelter from the fallout, the warning was received hours after the radiation had already settled. Scientists compelled the Atomic Energy Commission to send an emergency film that assured the public they were safe. In the film, AEC failed to explain preventative measures to avoid radiation exposure, such as consuming already exposed food or taking shelter during the next test.
The following years after the tests, "leukemia rates in Downwind Utah Mormon communities exceeded their non-Downwind counterparts by approximately five times, and pockets of thyroid cancer, especially in young children, spiked in areas which had never experienced childhood cancers." (AHF). The statistics led to an uproar from nearby families, demanding the shutdown of atmospheric tests. Families and fellow citizens were angered by the lack of precaution offered to them and the downplay of this radioactive fallout. The change was needed, and soon it was delivered.

Shut Down:

After many demands and pleas from scientists and the public for the shutdown of atmospheric testing of nuclear bombs, government officials began to listen and make changes. With proper protocol and safety measures, the first successful and fully contained underground nuclear test, the *Rainier Shot*, occurred at the Nevada Test Site in 1957. In 1958, President

Eisenhower announced self-imposed testing moratorium with the Soviet Union. There were no nuclear tests being conducted until 1961, when the Soviet Union announced its intentions to break the agreement and conducted over thirty-one nuclear tests in three months. The U.S. responded by resuming its nuclear testing program. In October 1962, the Cuban Missile Crisis brought the United States and Cuba to a standoff. The world was on the brink of starting a nuclear war. After the Soviets agreed to remove their nuclear weapons from Cuba, the crisis subsided, and efforts to efficiently continue testing resumed. In 1963, Great Britain, the United States, and the Soviet Union officially signed the Limited Test Ban Treaty. This treaty banned testing nuclear weapons in the atmosphere, outer space, and underwater. Scientists and government officials decided to move testing underground to continue the research of nuclear weapons. The urge to continue research was greatly influenced by the current state of the Cold War, as many countries were beginning to test their nuclear weapons in case of need during the war. As this was a new "playing field" for these weapons, scientists began more research, and government officials took further precautions to continue successfully and safely. To monitor the safety precautions at the site, the government created a safety group known as the RAD Safe Team to keep workers out of high radiation levels. Scientists at the Nevada Test Site conducted two types of testing; one type took place in the tunnels to monitor the effects of nuclear explosions on military hardware, and the other type took place in vertical holes in the valley to test new nuclear weapon designs (NATM).

Due to the new environment, unfortunate incidents occurred during the underground tests, such as the Baneberry incident in 1970. High water content in the detonation area created a fissure resulting in 300ft after the detonation. Radioactive particles filled the air and rained down all over the site, resulting in the death of two workers from leukemia due to exposure to

Iodine-131. This unfortunate incident halted underground testing for six months to form a proper investigation. The investigation discovered the high water content, creating a new standard for all nuclear test sites to go through appropriate geological analysis before testing. (Edwards). The U.S. continued underground testing for another 20 years, with the last underground nuclear weapons test, *Divider*, at the Nevada Test Site on Sept. 23, 1992.

Conclusion (Where the Nuclear Testing Program is at Today):

Many factors, including the end of the Soviet Union in December 1991, influenced the termination of traditional nuclear weapons testing. As the need for testing was not as imperative as it used to be during the Cold War, government officials worldwide agreed to make an effort to end nuclear weapons testing. The U.S. entered into a self-imposed moratorium on nuclear weapon testing in 1992. The United States, along with China, France, the United Kingdom, and Russia, signed the Comprehensive Nuclear-Test-Ban Treaty in 1996, which bans nuclear testing everywhere on the planet — surface, atmosphere, underwater and underground. Although the treaty has not been officially enforced, effort to ban nuclear weapon testing continues. In 2000, Congress created the National Nuclear Security Administration (NNSA) as a semi-autonomous agency within the U.S. Department of Energy. It is responsible for enhancing national security through the military application of nuclear science. The NNSA is also responsible for maintaining the Stockpile Stewardship, created after the United States ceased underground testing. The Stockpile Stewardship maintains U.S. nuclear weapons' safety, security, and reliability without full-scale testing. In 2010, the Nevada Test Site was renamed the Nevada

National Security Site, as the U.S. no longer needed it for testing and now for national security needs.

To help those who have been affected by the fallout, the government created the Radiation Exposure Compensation Act in 1990 to make payments to some of the people who claimed to have been affected by the fallout from the nuclear tests, but these payments are only available under a limited set of conditions. With the end of traditional nuclear weapons testing, it is essential to note that "although debate about the impact of low doses of radioactivity continues, a study completed in 2001 by the National Cancer Institute-Center for Disease Control tentatively concluded that exposure to fallout from nuclear tests during the 1950s was associated with an increase in U.S. cancer mortalities; fallout exposure may have caused more than 11,000 deaths" (Evans). As there is an end to traditional nuclear weapons testing, scientists will continuously study all radiation-related health concerns to prevent future cases.

References:

Ball, Howard. *Justice Downwind: America's Atomic Testing Program in the 1950's*. Oxford University Press, 1988.

Brooks, Antone L., et al. "How the Science of Radiation Biology Can Help Reduce the Crippling Fear of Low-Level Radiation." *Health Physics*, vol. 124, no. 5, 2023, pp. 407–424, https://doi.org/10.1097/hp.0000000000001677.

Edwards, Elisabeth. "Here's How the Baneberry Test Disaster Forever Changed Underground Nuclear Testing." *Warhistoryonline*, 4 Oct. 2022, www.warhistoryonline.com/cold-war/baneberry-test-operation-emery.html.

Evans, Michael. “The Making of the Limited Test Ban Treaty, 1958-1963.” Edited by William Burr and Hector L. Montford, *The Making of the Limited Test Ban Treaty, 1958-1963*, nsarchive2.gwu.edu/NSAEBB/NSAEBB94/#5.

Lane, Taylor. “Nevada National Security Site through the Years - Photos.” *Journal*, 17 Aug. 2022, www.reviewjournal.com/local/local-nevada/nevada-national-security-site-through-the-years-photos-2624752/.

Listwa, Dan. “Hiroshima and Nagasaki: The Long Term Health Effects.” *K=1 Project*, 9 Aug. 2012, k1project.columbia.edu/news/hiroshima-and-nagasaki.

National Atomic Testing Museum. “Oral History RAD Safe Team Panel Disscusion.” *YouTube*, YouTube, 28 June 2023, https://www.youtube.com/watch?v=-zuK6R9CICs.

Ropeik, David. “How the Unlucky Lucky Dragon Birthed an Era of Nuclear Fear.” *Bulletin of the Atomic Scientists*, 28 February 2018, thebulletin.org/2018/02/how-the-unlucky-lucky-dragon-birthed-an-era-of-nuclear-fear/.

“Castle Bravo.” *Nuclear Museum*, ahf.nuclearmuseum.org/ahf/history/castle-bravo/.

“Downwinders and the Radioactive West.” *PBS*, 3 Oct. 2022, www.pbs.org/video/downwinders-and-the-radioactive-west-usugap/.

“Hiroshima and Nagasaki: 75th Anniversary of Atomic Bombings.” *BBC News*, 8 Aug. 2020, www.bbc.com/news/in-pictures-53648572.

"Nevada Test Site Downwinders." *Nuclear Museum*, ahf.nuclearmuseum.org/ahf/history/nevada-test-site-downwinders/.